1. Population vs. Sample (Data Types)

- **Deep Explanation:** * **Population (N):** It is the complete collection of all possible observations. Imagine you want to find the average height of every person in India—that entire group is the population. Since it's often impossible to reach everyone, we use census methods.
    - **Sample (n):** It is a representative subset of the population. We use sampling because of time, cost, and feasibility constraints.
    - **Key Insight:** For a sample to be "good," it must be **Random** and **Unbiased**. If you only measure the height of basketball players, your sample won't represent the whole country.

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2. Measures of Central Tendency (Mean, Median, Mode)

- **Mean:** The arithmetic average. It is highly sensitive to **Outliers**.
    - *Example:* If 4 people earn $10k and 1 person earns $1 Million, the Mean will suggest everyone is rich, which is misleading.
- **Median:** The exact middle point. It divides the data into two equal halves (50th percentile).
    - *Example:* In the salary example above, the Median would stay near $10k, giving a truer picture of the group.
- **Mode:** The most frequent value. Essential for non-numerical data like "Which car model sells the most?"

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3. Variance and Standard Deviation (Measure of Dispersion)

- **Variance:** It measures how far each number in the set is from the mean. We square the differences so that negative distances don't cancel out positive ones.
- **Standard Deviation:** It is the square root of Variance. It brings the unit back to the original scale.
- **Real-world Use:** If you are buying a stock, a **High SD** means high risk/volatility, while a **Low SD** means stability.

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4. Normal Distribution (Gaussian Distribution)

- **Deep Explanation:** It is a perfectly symmetrical distribution. The Mean, Median, and Mode all lie at the peak.
- **Empirical Rule (68-95-99.7):**
    - 68% of data falls within 1 Standard Deviation.
    - 95% of data falls within 2 Standard Deviations.
    - 99.7% of data falls within 3 Standard Deviations (Anything outside this is often considered an **Outlier**).

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5. Standard Normal Distribution (Z-Score)

- **Concept:** This is a special Normal Distribution where **Mean = 0** and **SD = 1**.
- **Standardization:** It is the process of converting raw data into a Z-score.

- **Why?** In Machine Learning (like KNN or SVM), if one feature is 'Age' (0-100) and another is 'Salary' (0-1,000,000), the model will ignore Age because the Salary numbers are bigger. Standardization puts them on the same playing field.

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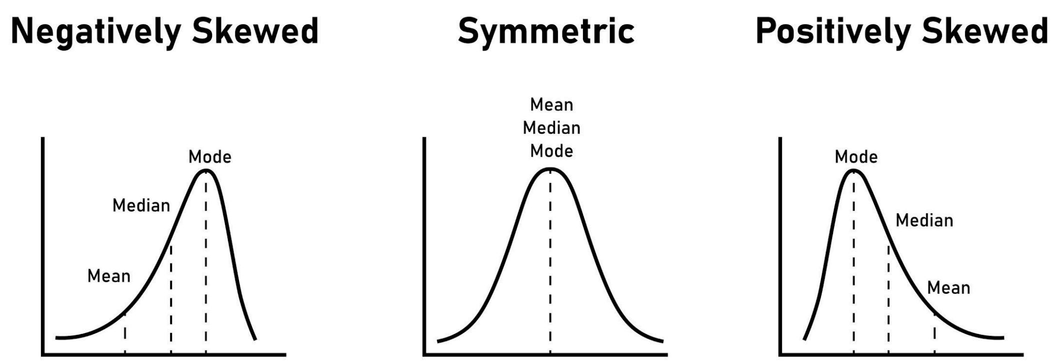
6. Skewness (Asymmetry of Data)

- **Positive Skew (Right-Skewed):** The tail is on the right. Most data is on the left. (Example: Household income, where most earn less, and a few earn billions). **Mean > Median**.
- **Negative Skew (Left-Skewed):** The tail is on the left. Most data is on the right. (Example: Age of death, most people die old, fewer die young). **Mean < Median**.



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7. Hypothesis Testing (The Logic of Science)

This is used to decide if a result is "Statistically Significant" or just happened by luck.

- **Null Hypothesis :** The "Status Quo" or "No effect" statement (e.g., "This new medicine doesn't work").ss
- **Alternate Hypothesis :** What you want to prove (e.g., "This medicine works").
- **Tests:**
  - **Z-test/T-test:** Compare means of two groups.
  - **ANOVA:** Compare means of 3 or more groups.
  - **A/B Testing:** Used by companies like Netflix/Amazon to test which button color or layout gets more clicks.